

USE CASES

What You Can Build with WAFT

1. Evolutionary AI Development

Breed Better Agents

Goal: Develop AI agents that improve through evolution

How:

```
waft new my_ai_lab
waft spawn --agent BaseAgent
waft evolve --generations 100
waft dashboard # Monitor progress
```

Output: Evolved agents with improved Scint handling, tracked lineage, and fitness metrics

2. Corporate Simulation

Simulate Business Operations

Goal: Model corporations with economics, employees, and departments

How:

```
from waft.core.corporations import CorporationsSystem

corps = CorporationsSystem()
corp = corps.create_corporation(
    name="My Startup",
    initial_capital=Decimal("1000000"),
)
corp.hire_employee(being_id, role="Engineer")
simulator.advance(months=12)
```

Output: Financial projections, runway analysis, organizational structure

3. Worldbuilding

Create Rich Fictional Universes

Goal: Build detailed worlds with characters, organizations, and lore

How:

- Create Realms with specific physics/rules
- Spawn Beings with personalities and goals
- Simulate interactions and history
- Generate documentation automatically

Output: Complete worldbuilding documents, character profiles, timeline

4. Document Generation

Professional PDF Production

Goal: Generate high-quality documents from data

Templates Available:

- Reports
- Briefs
- Memos
- Invoices
- Papers
- Manuals
- Guides
- Storybooks
- Character Sheets

Output: Professional PDFs via Typst, HTML, or LaTeX

5. D&D Campaign Management

Run and Track RPG Campaigns

Goal: Manage D&D 5e campaigns with AI assistance

Features:

- Character sheet generation
- Encounter creation
- Lore building
- Party state tracking
- Quest management

Output: Campaign materials, session notes, character documents

6. Epistemic Research

Track Knowledge and Learning

Goal: Quantify what AI (and humans) know and learn

Empirica Features:

- 13 epistemic vectors
- Preflight/Postflight assessment
- Learning quantification
- Gamification (XP, Karma)

Output: Session data, learning metrics, knowledge graphs

7. Multi-Agent Simulation

AI Town & Agent Coordination

Goal: Simulate multiple agents interacting

AI Town Features:

- Agent spawning and management
- Conversation system
- Voting mechanisms
- Memory sharing

Output: Agent interaction logs, emergent behaviors, town state

8. Self-Improving Systems

Build Systems That Fix Themselves

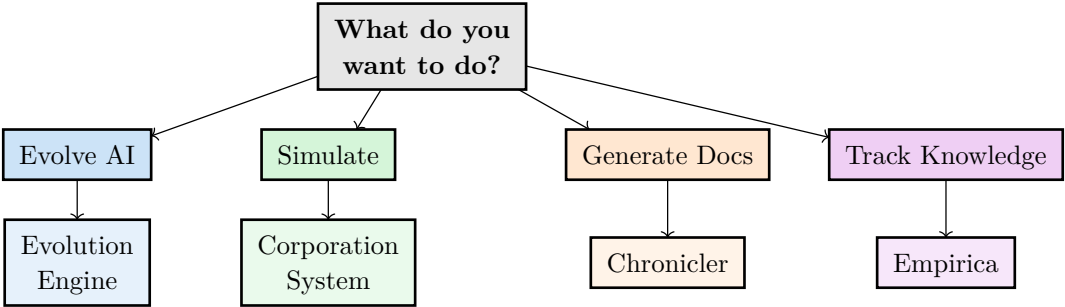
Goal: Create systems that detect and solve their own problems

Self-Engineering Modules:

- Problem Detector
- Diagnostician
- Solution Engineer
- Self Modification
- Iteration Loop

Output: Self-improving code, diagnostic reports, fix logs

Use Case Decision Tree



Quick Start by Use Case

Use Case	Command
Evolve Agents	<code>waft new lab && waft evolve</code>
Create Corporation	<code>waft corp create "My Startup"</code>
Generate Brief	<code>waft chronicle brief --data data.json</code>
Start Session	<code>waft session create</code>
Spawn Being	<code>waft being spawn --reality my_reality</code>
Run D&D	<code>waft quest create --scenario dungeon</code>

